

# Description Logic

## ➤ $\top$ : Top Concept (Universal Concept)

- ▶ Represents the set of ALL individuals in the domain. Every individual belongs to  $\top$ .
- ▶  $\top^I = \Delta^I$  (the entire domain)
  - ▶  $\text{Person} \sqsubseteq \top$  (every Person is a thing)
  - ▶  $\exists \text{owns}.\top$  (owns at least one thing (of ANY type))
  - ▶  $\forall \text{teaches}.\top$  (teaches only things (no restriction – always true))

## ▶ $\perp$ : Bottom Concept (Empty Concept)

- ▶ Represents the empty set – no individual can ever belong to  $\perp$ . It signals a logical contradiction.
- ▶  $\perp^I = \emptyset$  (empty set. nothing belongs here)
- ▶  $\text{Student} \sqcap \neg \text{Student} \equiv \perp$  (impossible to be both Student and not Student)

# Reasoning using DL

## ► Subsumption Checking

- **Question:** Is concept C more general than concept D? (Is  $D \sqsubseteq C$ ?)
- **TBox:**
  - $\text{PhDStudent} \equiv \text{Student} \sqcap \exists \text{hasAdvisor. Professor}$
  - $\text{GradStudent} \equiv \text{Student}$
- **Query:**  $\text{PhDStudent} \sqsubseteq \text{GradStudent}$ ?
- **Answer:** YES, because PhDStudent requires being a Student, which is exactly GradStudent

## ► Satisfiability Checking

- **Question:** Can this concept have any instances (is it logically consistent)?
  - $\text{Student} \sqcap \neg \text{Person}$
- **TBox says:**  $\text{Student} \sqsubseteq \text{Person}$
- So  $\text{Student} \sqcap \neg \text{Person} \equiv \perp$  (empty set)  $\rightarrow$  UNSATISFIABLE

# Satisfiability Checking

- ▶ **Question:** Can a concept ever have any instances? Or is it logically impossible (equivalent to  $\perp$ )?
- ▶ **TBox:**
  - ▶  $\text{Student} \sqsubseteq \text{Person}$
  - ▶  $\text{Professor} \sqsubseteq \text{Person}$
- ▶ **Concept to check:**  $X \equiv \text{Student} \sqcap \neg \text{Person}$
- ▶ **Is X satisfiable?**
- ▶ **Reasoning:**
  - ▶ Step 1: X requires being a Student ( $\text{Student} \sqsubseteq \text{Person} \rightarrow \text{must be a Person}$ )
  - ▶ Step 2: X requires NOT being a Person ( $\neg \text{Person}$ )
  - ▶ Step 3: Contradiction; must be Person AND not Person
- ▶ **Result:**  $X \equiv \perp$  (UNSATISFIABLE)

# Satisfiability Checking

- ▶ TBox:
  - ▶  $\text{ChildPatient} \sqsubseteq \text{Patient}$
  - ▶  $\text{ChildPatient} \sqsubseteq \leq 0 \text{ prescribedBy. Physician}$  (no physician prescribes to child directly)
- ▶ Concept to check:  $Y \equiv \text{ChildPatient} \sqcap \exists \text{ prescribedBy. Physician}$
- ▶ Is Y satisfiable?
- ▶ Reasoning:
  - ▶ Step 1: Y requires at least 1 prescribing Physician ( $\exists \text{ prescribedBy. Physician}$ )
  - ▶ Step 2: ChildPatient allows AT MOST 0 physicians ( $\leq 0 \text{ prescribedBy. Physician}$ )
  - ▶ Step 3: Cannot have  $\geq 1$  AND  $\leq 0$  simultaneously
- ▶ Result:  $Y \equiv \perp$  (UNSATISFIABLE)

# Satisfiability Checking

- ▶ Concept to check:  $Z \equiv \text{Student} \sqcap \text{Employee}$
- ▶ Is Z satisfiable?
- ▶ Reasoning:
  - ▶ No TBox rule says Student and Employee are disjoint.
  - ▶ An individual can be both.
- ▶ Result: Z is SATISFIABLE (e.g., a part-time working student)

# Subsumption

- **Question:** Is concept C more specific than concept D?
- ▶ Is every C necessarily a D?
- ▶ C is subsumed by D (written  $C \sqsubseteq D$ ) if in every interpretation,  $C^I \subseteq D^I$ .
- ▶ **TBox:**
  - ▶  $\text{PhDStudent} \sqsubseteq \text{Student} \sqcap \exists \text{hasSupervisor. Professor}$
- ▶ Does  $\text{PhDStudent} \sqsubseteq \text{Student}$  hold?
- ▶ Reasoning:
  - ▶ PhDStudent requires being a Student (from the definition).
  - ▶ Therefore every PhDStudent is necessarily a Student.
- ▶ Result:  $\text{PhDStudent} \sqsubseteq \text{Student}$

# Indirect subsumption (chain):

- ▶ **TBox:**
  - ▶  $\text{PhDStudent} \equiv \text{Student} \sqcap \exists \text{hasSupervisor. Professor}$
  - ▶  $\text{Student} \sqsubseteq \text{Person}$
- ▶ Does  $\text{PhDStudent} \sqsubseteq \text{Person}$  hold?
- ▶ Reasoning:
  - ▶  $\text{PhDStudent} \sqsubseteq \text{Student}$  (from definition)
  - ▶  $\text{Student} \sqsubseteq \text{Person}$  (from TBox axiom)
  - ▶ Therefore  $\text{PhDStudent} \sqsubseteq \text{Person}$  (by transitivity)
- ▶ Result:  $\text{PhDStudent} \sqsubseteq \text{Person}$



# Subsumption does NOT hold:

- ▶ **TBox:**
  - ▶  $\text{Doctor} \equiv \text{Person} \sqcap \exists \text{treats.Patient}$
  - ▶  $\text{Student} \sqsubseteq \text{Person}$
- ▶ Does  $\text{Student} \sqsubseteq \text{Doctor}$  hold?
- ▶ Reasoning:
  - ▶ Student requires being a Person
  - ▶ Student does NOT require treating any Patient
- ▶ Result:  $\text{Student} \sqsubseteq \text{Doctor}$  does NOT hold (a student is not necessarily a doctor)